# Lab Experiment - 4

## Aim

To implement a Finite State Machine (FSM) for morphological parsing in Python to generate plural forms of English nouns.

## Algorithm

1. Start the program.
2. Read a singular noun from the user.
3. Check the ending of the noun.
4. If the noun ends with `s`, `sh`, `ch`, `x`, or `z`, add `es`.
5. If the noun ends with a consonant followed by `y`, replace `y` with `ies`.
6. Otherwise, add `s`.
7. Display the plural form.
8. Stop the program.

## Python Program

```
def plural(noun):
    state = "START"

    if noun.endswith(("s", "sh", "ch", "x", "z")):
        state = "ADD_ES"
        result = noun + "es"

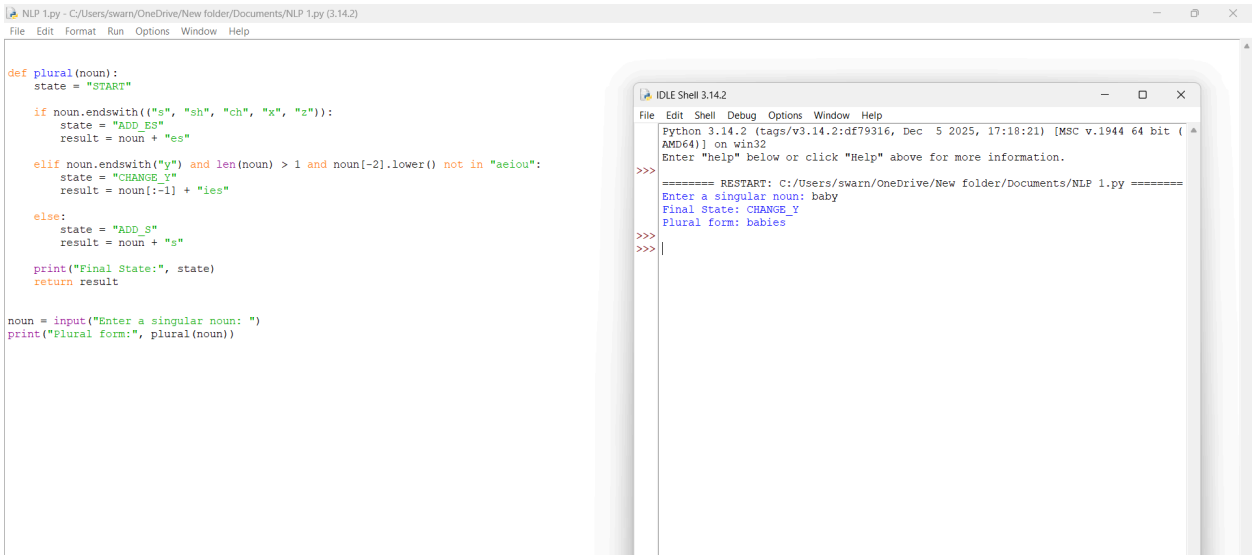
    elif noun.endswith("y") and len(noun) > 1 and noun[-2].lower() not in "aeiou":
        state = "CHANGE_Y"
        result = noun[:-1] + "ies"

    else:
        state = "ADD_S"
        result = noun + "s"

    print("Final State:", state)
    return result


noun = input("Enter a singular noun: ")
```

print("Plural form:", plural(noun))




# Result

Thus, the Finite State Machine-based morphological parsing program was implemented successfully to generate plural forms of English nouns using Python.